MODULE 21 · DOMAIN 3 - IMPLEMENT COMPUTER VISION SOLUTIONS

Generating and editing images

GPT-image models, generation controls, reference images, masks and inpainting, input fidelity, and filtering errors.

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Generation starts from **words**. Editing starts from **pixels you want to keep**. Every editing control decides how much of the original survives.

Image models in Foundry

MODEL	ACCESS	BEST FOR
GPT-Image-2.5 Sunburst	Generally available	Most capable; editing precision
GPT-Image-2.5 Flare	Generally available	Fastest high-quality everyday generation
GPT-Image-2	Generally available	High resolution, 4K, flexible aspect ratios
GPT-Image-1.5 and 1	Limited access	Realism and instruction following
GPT-Image-1-Mini	Limited access	Cheap, fast bulk drafts; no face preservation

HONEST LIMITS

DALL-E 3 retired on 4 March 2026, and existing deployments stopped working. Older answers naming it are out of date.

Images API or Responses API?

IMAGES API

One request, one result

- A generations endpoint and an edits endpoint
- Direct control of every parameter
- Suits batch jobs and app features

RESPONSES API

Images inside a conversation

- Image generation as a tool the model calls
- Refine an image over several turns
- Suits assistants and agents

ON THE EXAM

The Images playground in the Foundry portal has View code, which gives Python and cURL for your exact settings.

What every generation request returns

- The **model** parameter is your **deployment name**, as with every Azure OpenAI call.
- GPT-image models return **base64 data only**, in b64_json. There's **no URL option**.
- **response_format isn't supported** for GPT-image models.
- Request **1 to 10** images at once, with the **n** parameter.
- Expect **10 to 30 seconds** per request. The default quota is **5 images a minute**.

> THE SIGNAL
>
> Code migrated from DALL-E that reads a URL breaks on GPT-image models. Decode b64_json instead.

Parameters you can set

PARAMETER	OPTIONS	NOTES
`size`	1024x1024, 1024x1536, 1536x1024	GPT-Image-1 series; square is fastest
`size (GPT-Image-2)`	Custom, edges multiples of 16	Long edge up to 3,840; ratio up to 3:1
`quality`	low, medium, high	Lower is faster; default high
`output_format`	png (default) or jpeg	WebP isn't supported in Azure
`background`	auto or transparent	Transparent needs PNG output
`stream, partial_images`	true; 1 to 3 partial images	Faster visual feedback

ON THE EXAM

output_compression, from 0 to 100, shrinks JPEG files. Transparency and JPEG can't be combined.

Generating an image

```
result = client.images.generate(
    model="gpt-image-2",
    prompt="Flat vector sticker of a smiling coffee cup, thick outline",
    size="1024x1024",
    quality="medium",
    output_format="png",
    background="transparent",
    n=2)

for i, image in enumerate(result.data):
    with open(f"sticker_{i}.png", "wb") as f:
        f.write(base64.b64decode(image.b64_json))
```

ON THE EXAM

PNG plus transparent gives a cut-out sticker. The images arrive as base64, decoded before saving.

The edits endpoint is different

GENERATIONS

Words in, image out

- JSON request body
- A prompt and controls
- No input image

EDITS

Images in, image out

- multipart/form-data, not JSON
- Up to 16 input images, each under 50 MB
- PNG or JPG inputs, plus an optional mask

ON THE EXAM

Sending an edit request as JSON is a common first mistake. Files travel as form data.

Three kinds of image edit

- **Prompt-driven modification:** an image plus an instruction, with no mask. 'Add a beach ball in the centre.'
- **Reference images:** several inputs combined, such as a product photo and a style example.
- **Mask-based inpainting:** change only a marked region, and leave the rest untouched.
- Newer models can add or remove one aspect of an image while keeping everything else.
- Choose the lightest edit that works: a mask adds precision, but also preparation.

ON THE EXAM

The study guide names all three: prompt-driven modifications, reference media, and mask-based edits.

The rules for a mask

- **Fully transparent pixels**, with an alpha of zero, mark the area **to edit**.
- **Opaque pixels** mark the area to **keep**.
- The mask must be a **PNG**, because JPEG has no transparency.
- The mask must have the **same dimensions** as the input image.
- The prompt describes what should appear in the transparent area.

> WATCH FOR THIS
>
> Painting the area black or white doesn't work. Only transparency marks where to edit.

Transparent means change this. **Opaque** means keep this. The mask is a **PNG**, the same size as the image.

Input fidelity

- **input_fidelity** controls how closely the output matches the input's **style and features**.
- **High** preserves **faces** more accurately, for subtle edits that leave unrelated areas alone.
- **GPT-Image-1-Mini doesn't support it**, and has no dedicated face preservation.
- GPT-Image-1.5 and GPT-Image-2 support high input fidelity.
- Use it where **faces or likeness** matter, such as edits of real people or products.

> THE SIGNAL
>
> 'Faces change when we edit portraits' points to input fidelity on a model that supports it, not a stronger prompt.

Inpainting with a mask

```
result = client.images.edit(
    model="gpt-image-2",
    image=open("storefront.png", "rb"),
    mask=open("sky_mask.png", "rb"),
    prompt="Replace the sky with a warm sunset, keep the shop unchanged",
    input_fidelity="high",
    size="1536x1024")

edited = base64.b64decode(result.data[0].b64_json)
```

ON THE EXAM

sky_mask.png is a PNG, the same size as storefront.png, with the sky fully transparent.

Requirement to control

REQUIREMENT	CONTROL
Stickers or logos on any background	background transparent, output_format png
Small files for web thumbnails	output_format jpeg, with output_compression
Users waiting on a slow image	stream, with partial_images
A wide 4K banner	GPT-Image-2, custom size
Hundreds of cheap draft concepts	GPT-Image-1-Mini, quality low
Change only one region	A PNG mask, same size, area transparent
Keep a person's face in an edit	input_fidelity high, on a supporting model

ON THE EXAM

The study guide skill is choosing controls. Most wrong answers pick a real control for the wrong requirement.

Filtering and common errors

- A flagged **prompt** returns error code **contentFilter**: the task failed on safety grounds.
- A flagged **output** returns a different message: the generated image was filtered.
- **Photorealistic images of minors are blocked by default**, unless access is approved.
- **429:** quota exceeded, so back off or request more. **401** with managed identity: add Cognitive Services OpenAI User.
- **Timeouts:** stream, simplify the prompt, or request a smaller size.

> ON THE EXAM
>
> The two filter messages tell you whether to change the prompt or accept that the result itself was unsafe.

> A team migrates an app from DALL-E 3 to a GPT-Image-1.5 deployment. The code sets response_format to 'url', then downloads the image from data[0].url.
>
> Every request now fails, even though the deployment name and credentials are correct.

What should the team change?

A Keep response_format as 'url' and wait for the URL to become available.

B Switch the application back to its existing DALL-E 3 deployment.

C Request a higher image quota for the new GPT-Image-1.5 deployment.

D Remove response_format, and decode the b64_json field instead.

◆ PAUSE AND DECIDE

What should the team change?

A. Keep 'url' and wait.
GPT-image models have no URL option at all. Waiting changes nothing.

B. Back to DALL-E 3.
Retired in March 2026. Existing deployments no longer work.

C. More quota.
Quota problems return 429. These requests fail on format, not volume.

D. Drop response_format; decode b64_json.
GPT-image models return base64 only, and don't support response_format.

THE SIGNAL

A DALL-E migration that reads URLs breaks for exactly one reason: GPT-image models only return base64.

> A retailer wants to replace only the sky in product photos. The team sends each photo with a JPEG mask, in which the sky is painted solid black.
>
> Results change the whole image, including the product itself.

What should the team change?

A. Set input_fidelity to high so the product stays closer to the original.

B. Describe the product in far more detail in the prompt so it is preserved.

C. Send a PNG mask, the photo's size, with the sky fully transparent.

D. Use the generations endpoint, and describe the photo in the prompt.

◆ PAUSE AND DECIDE

What should the team change?

A High input fidelity.

Helps preservation, but without a valid mask nothing marks the sky as the only area to change.

B More detailed prompt.

Words can't draw a boundary. The mask is what limits the edit.

C PNG mask, same size, sky transparent.

Meets every mask rule, so only the transparent sky is edited.

D Generations endpoint.

Creates a new image from words, discarding the real product photo.

THE SIGNAL

JPEG and 'painted black' are two broken mask rules. Transparency in a PNG is the only valid marker.

> A company edits real employee portraits to change outfits for a campaign. It uses a GPT-Image-1-Mini deployment to keep costs down.
>
> Faces noticeably change in most results, and setting input_fidelity has no effect.

What should the team do?

A Keep GPT-Image-1-Mini, and raise quality to high on every edit request.

B Use a model supporting input_fidelity, like GPT-Image-1.5, set high.

C Add 'keep the face exactly identical' to the start and end of each prompt.

D Set n to 10 on every request, and let a reviewer pick the closest face.

◆ PAUSE AND DECIDE

What should the team do?

A. Mini at high quality.
Sharper images, but still no input fidelity or face preservation.

B. A supporting model, input_fidelity high.
High input fidelity preserves faces; Mini doesn't support it at all.

C. Stronger prompt wording.
Asks for the outcome without the capability that produces it.

D. Ten attempts per edit.
Ten times the cost, and faces may still drift in all ten.

THE SIGNAL

'Input fidelity has no effect' plus GPT-Image-1-Mini: that model doesn't support it.

> A design tool generates icons that must sit on any website background. To keep files small, it requests JPEG output, with 'transparent background' in the prompt.
>
> Every icon arrives with a solid white background.

What should the team change?

A Set background to transparent, and switch output_format to png.

B Set output_compression to 0 so the JPEG keeps full image quality.

C Repeat 'transparent background' twice in the prompt, in capitals.

D Request WebP output, which supports transparency and small files.

◆ PAUSE AND DECIDE

What should the team change?

A background transparent, PNG.
The documented combination for a transparent background.

B No compression.
A higher-quality JPEG is still a JPEG, with no transparency.

C Stronger prompt.
Transparency is a parameter, not something a prompt can switch on.

D WebP output.
WebP isn't supported in Azure OpenAI in Foundry Models.

THE TIEBREAKER

Transparency is a parameter and a format: background transparent, output_format png. Neither alone works.

Images in five lines

- **GPT-image models return base64 only.** DALL-E 3 retired in March 2026.
- **Transparent background = background transparent + PNG.** No WebP in Azure.
- **Edits are form data:** up to 16 images, each under 50 MB.
- **Masks: PNG, same size, transparent where to edit.**
- **Input fidelity high** preserves faces, but not on GPT-Image-1-Mini.

> ON THE EXAM
>
> Next, module 22: generating and editing video with Sora 2.